

# Gyroscope Test Bed

## Base Information

Test Name: Gyroscope Test Bed

Subsystem: Hardware

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Date Executed: April 10th, 2026 - July 6th, 2026

For Discovery Days: April 28th, 2026 - May 1st, 2026

Location: University of Washington

Status: Completed

## Objective

The objective of this design is to create a gyroscope that has the degrees of freedom to rotate about the x, y, and x axis so that roll, pitch, and yaw capabilities are simulated about the drone. The gyroscope's purpose is to allow the software team to fine tune the PID controller by simulating disturbances in all degrees of freedom with the gyroscope.

## Related Documents

Link to Bearing Orders: [Current Order](#) | [McMaster-Carr](#)

Design Model Website: [3D Flight Lab – 3DFlightlab](#)

Gyroscope CAD Folder: [WAAR\\_Gyro\\_Iteration01and02\\_2026\\_ErikaAndSkye](#)

## Equipment

- Ball Bearing, 0.25" diameter (x 4)
- Needle-Roller Thrust Bearing with 2.5" diameter (x 1)
- 0.032" Thick Washer for 2.5" Shaft Diameter Needle-Roller Thrust Bearing (x 2 )

- PLA for 3D printing
- PVC pipe with 0.5 inch diameter
- Add screws

## Requirements and Constraints

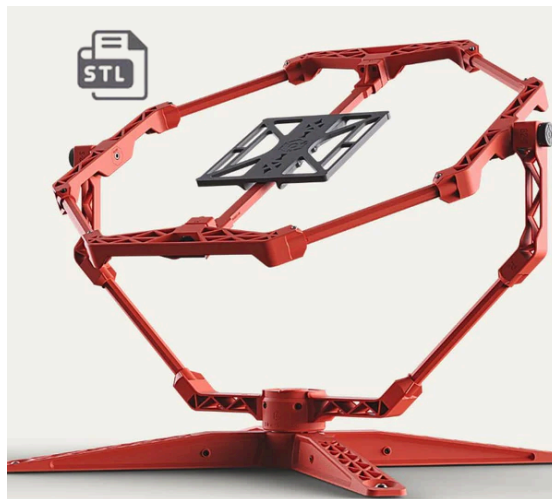
| Requirement         | Value                                                                                      | Notes                                                                    | Status |
|---------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------|
| Degrees of Freedom  | = 3 DOF                                                                                    | Must have pitch, roll and yaw capabilities.                              | MET    |
| Safety Requirements | The design must create a cage around the drone, and the drone must be rigidly mounted.     | Prevent user injury and damage during operation.                         | MET    |
| Load Capabilities   | The gyroscope should be able to withstand vibrations from the drone and rotational forces. | Ensures structural integrity under forces and vibrations from the drone. | MET    |
| Cost                | The gyroscope should be less than \$60.                                                    | Try to keep bearings under \$5 each.                                     | MET    |

## Design Idea

The gyroscope will look similar to the design as seen below in Figure 1, with an octagonal top structure that has two degrees of freedom and four ball bearings, and a base that has one degree of freedom and two thrust bearings. The drone will be attached to the drone support frame with ideally four velcro straps or zip ties.



Figure 1: Image of the inspiration behind the model for the gyroscope.



## Design Procedure

### Overview

Following the design of the previously shown image to create a fully 3D printed gyroscope utilizing the cheapest and quickest methods. Time is of the essence and cost is a factor. Bearings were selected to fit a tight cost criteria and the design will revolve around them. If the budget allows, larger thrust bearings can be purchased to better support the weight of the drone and prevent wobbling. The base stand, base, base connectors, vertical connectors, and four components of the octagon were designed in SolidWorks. The thin circular rods that connect each of the components was not 3D printed but made of PVC. No bearing caps were needed like in the image, as the ball bearings were designed to fit in a small cut extrude feature. The base stand was made to be a shelled component, that was filled with sand for weight.

### **Vertical Connector to Octagon**

The gyroscope was designed to connect the vertical connector with one side of the octagon, and allow rotation at that point with a ball bearing. The side of the octagon had a hole in it with which the bearing fit into, and the vertical connector had an adapter on it's side which fit into the inner diameter of the bearing. See images below for clarification, where Figure 2 shows how the bearing fit into one side of the octagon, Figure 3 shows a close up of the rotational joint, and Figure 4 shows the entire assembly.

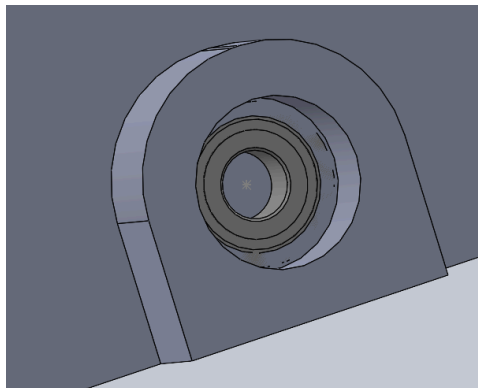


Figure 2: Bearing in extrude cut feature.

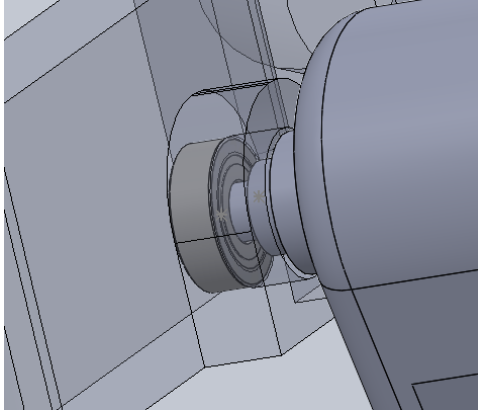


Figure 3: Close up image of connection between components.

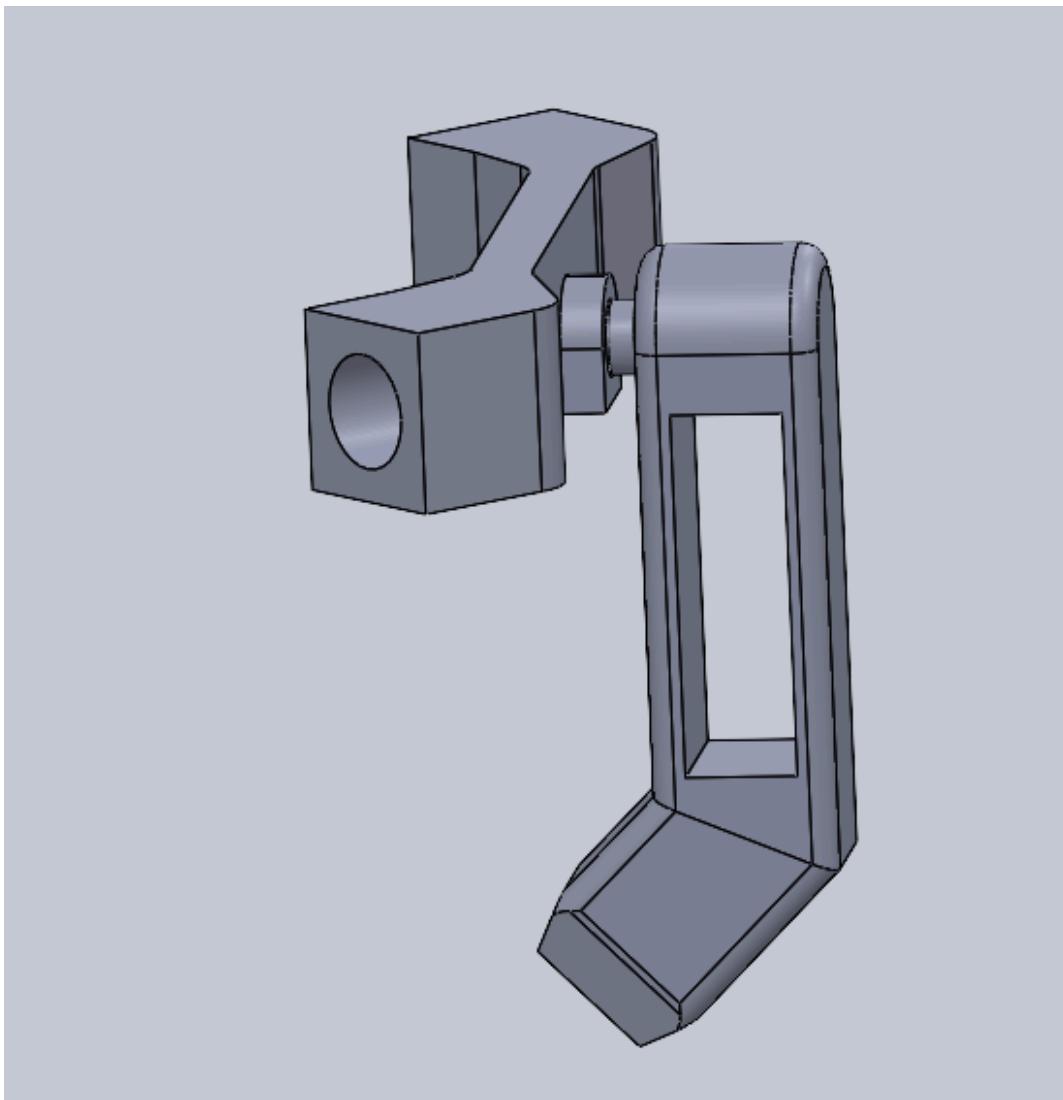


Figure 4: Image of the assembly.

## Base Design

The base was designed so that with weight-bearing from the loaded payload, the base would compress slightly. That is why there are slight gaps between the components. The vertical connector slots in from the side and a pin is pushed in from the top to secure it in place. Two thrust bearings of different sizes are used to allow for proper rotation of the base, while minimizing friction. A nut is screwed from the top through the center of the base to secure all the vertically stacked components together. When assembled and tested there was a slight wobble to the base when rotating about the z-axis, due to less compression of the components than initially predicted when loaded. However, adding a second thrust bearing and making other modifications reduced the friction enough that three degrees of freedom were achieved for testing purposes.

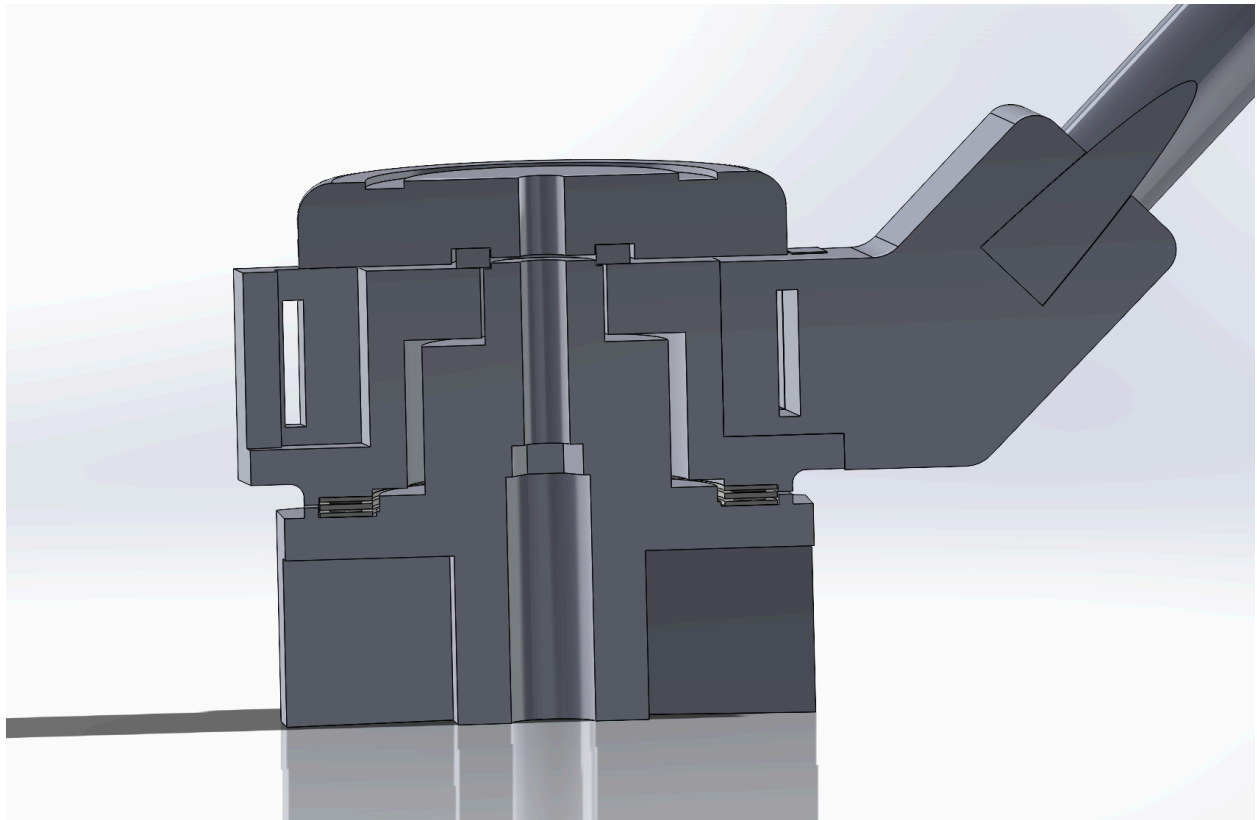


Figure 5: ...

## Issues & Iteration Improvement

## **First Iteration Issues**

- Cracking noticed around dovetail connections on base
- Base connectors slightly too high
- Excessive friction on thrust bearing connection
- Existing PVC connections required glue to be stable, tolerance is too great
- Ball bearing connection needed an adapter to function
- Some warping of components from 3D printing

## **Second Iteration Issues and Fixes**

- Some wobbling in the base when there is rotation in multiple degrees of freedom
- Some tolerance issues in the PVC connections, which were fixed by printing adapters
- Drone support frame was outdated and had to be re-designed
- PVC attaching from the base to the vertical connectors was too long, and trimmed by 0.3 inches on each side
- Screws tended to pull up material from the components, and so were often screwed in partially
- The gyroscope was balanced properly by screwing on some washers to the octagon

## **Second Iteration Improvements**

- Base (with thrust bearing) rotated much more smoothly than before, and reduced previous wobbling a little
  - The base was completely re-designed and went from having one to two bearings
- Base supports were successfully re-designed to prevent cracking
- Ball bearing connection/socket was re-designed to have a tight fit and allow easy rotation

- Material was reduced on components in the octagon while maintaining stability
- Strengthened weakened areas with more material or by re-designing the area
- Redesign PVC mounting to use screws or some other method that eliminates the need for glue

## Conclusions

The first iteration of the gyroscope had substantial issues that prevented functionality in all three degrees of freedom. The base had a lot of friction while rotating, and the ball bearings did not connect well with the proper components. PVC connections required glue to be stable and there were a few issues with tolerancing. In the second iteration there were also problems that came up, but those issues had easy fixes and generally didn't influence the functionality of the gyroscope. Balancing the gyroscope involved screwing on a few washers on one side of the octagon, and there were issues with accurately cutting the PVC to the intended size as a hand saw was used. Due to small discrepancies in the length, some of the PVC had to be trimmed down. However, in general the base rotated much better than before, and by using screws instead of glue the entire gyroscope was much more stable. The finished product was able to be used to fine tune the PID controller, and will ideally last for another year.